13.Implement the Car Price Prediction Model using Python

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import OneHotEncoder

from sklearn.compose import ColumnTransformer

from sklearn.pipeline import Pipeline

from sklearn.metrics import mean_squared_error, r2_score

data = {

  "Year": [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],

  "Present_Price": [5.5, 6.0, 7.0, 8.5, 9.0, 10.0, 11.5, 13.0, 14.0, 15.0],

  "Kms_Driven": [50000, 45000, 40000, 35000, 30000, 25000, 20000, 15000, 10000, 5000],

  "Fuel_Type": ["Petrol", "Diesel", "Petrol", "Diesel", "Petrol",

          "Diesel", "Petrol", "Diesel", "Petrol", "Diesel"],

  "Selling_Price": [3.5, 4.2, 5.0, 6.5, 7.0, 8.0, 9.5, 11.0, 12.0, 13.0]}

df = pd.DataFrame(data)

X = df.drop("Selling_Price", axis=1)

y = df["Selling_Price"]

categorical = ["Fuel_Type"]

preprocessor = ColumnTransformer(

  [("cat", OneHotEncoder(handle_unknown="ignore"), categorical)],

  remainder="passthrough")

model = Pipeline([

  ("preprocessor", preprocessor),

  ("regressor", LinearRegression())])

X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Actual Prices:", y_test.values)
print("Predicted Prices:", y_pred)
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
car = pd.DataFrame({
    "Year": [2022],
    "Present_Price": [12.0],
    "Kms_Driven": [18000],
    "Fuel_Type": ["Petrol"]})
print("Predicted Car Price:", model.predict(car)[0])
```

OUTPUT:

```
Actual Prices: [12.   4.2]
Predicted Prices: [12.  4.]
Mean Squared Error: 0.020000000000000746
R2 Score: 0.9986850756081525
Predicted Car Price: 10.000000000000002
```